

### Code-Layout, Readability and Reusability

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 15 March 2024 |
| <b>Team ID</b>       | XXXXXX        |
| <b>Project Name</b>  | XXXXXX        |
| <b>Maximum Marks</b> | 5 Marks       |

#### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

#### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                           |
|------|---------------------------|-----------------------------------------------|-------------------------------|-----------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | Makes the code easy to read.      |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | Easy to locate project files.     |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Improves code understanding.      |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Easy to maintain.                 |
| 5    | Code Comments             | Inline and block comments explain logic       | Partial                       | Important sections are explained. |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Simple to update in future.       |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | Code is clean.                    |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           | Prevents common runtime errors.   |

#### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology | Where Reused                                 | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|----------------------------------------------|-----------------------------------------|
| 1    | User Input Form         | HTML, Jinja2          | Collecting user questions and learning tasks | High                                    |
| 2    | AI Response Handler     | Python, Fast API      | Processing all AI requests and responses     | High                                    |

|   |                      |               |                                        |        |
|---|----------------------|---------------|----------------------------------------|--------|
| 3 | Web Interface Design | HTML, CSS     | Main application interface             | High   |
| 4 | API Configuration    | Python (.env) | Secure API key management              | Medium |
| 5 | Styling Components   | CSS           | Buttons, forms, cards, and page layout | High   |

Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments                                                                |
|--------------------------|--------------|-------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | Project files are organized in a clear and simple structure.            |
| Readability              | 5            | Code is easy to understand with meaningful names and proper formatting. |
| Reusability              | 4            | Common functions and templates can be reused in future updates.         |
| Documentation / Comments | 4            | Necessary comments are added to explain important parts of the code.    |
| Overall Score            | 4.5 / 5      | The project follows good coding practices and is easy to maintain.      |